

EM, OLI & TIMMONDS | FUTURES BRIEFING | OCTOBER 2026 · STORY SPINE: ONCE UPON A TIME

Three Worlds of 2041

How AI could reshape our industry, our customers and our planet over the next 15 years

THREE SCENARIOS, NOT THREE PREDICTIONS

On 6 April 2026, four people watched Earth set behind the Moon. Fifteen years from now, the Moon may be a place where people go to work. This is not a dream. It is three futures we can already see coming, and the choices that decide which one we get.

IN THIS BRIEFING today's signals, our method, three worlds of 2041, the odds, and what to do from Monday.



Mohammad

Senior Futures Consultant, Em, Oli & Timmonds · Part 1 | The Creative Solution

EARTH SETS BEHIND THE MOON, PHOTOGRAPHED BY THE ARTEMIS II CREW, 6 APRIL 2026. IMAGE: NASA

NEXT: the story map: how this briefing unfolds in fifteen steps.

One story, fifteen steps

THE STORY MAP | HOW THIS BRIEFING UNFOLDS · STORY SPINE: ONCE UPON A TIME

Our brief: imagine the world in 15 years' time, considering AI's impact on our industry and on the whole world, and decide what E,O&T should build now without betting on one forecast. The story uses an adapted, branching Story Spine: the three "Because of that" steps are alternative worlds. Facts and projections are tagged; the 2041 worlds are imagined scenarios.

SCOPE

Industry: strategic consulting | Clients: energy, logistics and digital services |
World: work, daily life, infrastructure and the space frontier

NEXT: WE START WITH THE FACTS: FOUR SIGNALS ALREADY VISIBLE IN 2026.

- 3 **EVERY DAY** — The future is leaking in — TOOL 2 of the 4 lenses
- 4 **BUT ONE DAY** — The Futures Engine — TOOL Seven creative tools
- 5 **BECAUSE OF THAT (1)** — Symbiosis 2041 — TOOL Moonshot, upward What if
- 6 **BECAUSE OF THAT (2)** — Overheat 2041 — TOOL Downward What if
- 7 **BECAUSE OF THAT (3)** — Patchwork 2041 — TOOL Tick-Tock
- 8 **ZOOM IN (1)** — From your door to orbit — TOOL Moonshot, exaptation
- 9 **ZOOM IN (2)** — The new gold — TOOL Tick-Tock, all 4 lenses
- 10 **UNTIL FINALLY** — Which branch are we on? — TOOL Possibility tree
- 11 **UNTIL FINALLY: WATCH** — Lights that change action — TOOL Contingency planning
- 12 **EVER SINCE: RESPOND** — Sell better decisions — TOOL Decision Passport
- 13 **STORY IN ONE WORLD** — Omar in eight steps — TOOL Story Spine, 5 Cs
- 14 **AND EVER SINCE THEN** — Back to Monday morning — TOOL Working backwards
- 15 **EVER SINCE: FIRST TEST** — The first 90 days — TOOL Working backwards, gates
- 16 **EVER SINCE: KEEP CREATING** — A creative routine — TOOL Safety, Octalysis, stimuli
- 17 **THE CARRY-OUT MESSAGE** — The Moon is closer than ever — TOOL 5 Cs of storytelling

The future is already leaking in

TODAY'S SIGNALS | 4 LENSES: HARNESSING TRENDS AND CHALLENGING ORTHODOXIES · STORY SPINE: EVERY DAY

415
TWh

used by data centres in 2024, about 1.5% of world electricity, with about 945 TWh by 2030 as a base-case projection. (IEA) — *Energy becomes AI's ceiling.*

54,000

debris objects larger than 10 cm circle Earth, plus about 1.2 million between 1 and 10 cm. (ESA) — *Orbit is a finite resource we can pollute.*

170M

new jobs created and 92M displaced by 2030 across major economic and technological trends, including AI; 39% of core skills change. (WEF) — *Skills, not jobs, become the scarce asset.*

APRIL
2026

Artemis II flies four astronauts around the Moon, the first crewed lunar flight since 1972. (NASA) — *Space moves from exploration to economy.*

Four signals we can measure in 2026, found with two of the 4 lenses of innovation: harnessing trends and challenging orthodoxies. Every world grows out of them.

Challenging orthodoxies. The old belief "AI is software" no longer holds. AI is an energy, materials and orbit industry, and that is exactly where our three worlds split apart.

NEXT: HOW DO WE TURN THESE SIGNALS INTO FUTURES? OUR FUTURES ENGINE.

The E,O&T Futures Engine

OUR METHOD | A MORE CREATIVE FORM OF SCENARIO PLANNING · STORY SPINE: BUT ONE DAY

How the signals become worlds. Seven tools work like gears, and each produces a visible output on the slides shown. We first ran a controlled What if (keep AI access fixed, change only energy headroom), then combined both uncertainties into three worlds.

- 1 **Harness trends** — Spot what is already moving — SLIDES 3, 9
- 2 **Counterfactual What if** — Change one driver, upward and downward — SLIDES 5, 6
- 3 **Possibility tree** — Branch it and put odds on every branch — SLIDES 10
- 4 **Tick-Tock** — Flip each assumption into its opposite — SLIDES 7, 9
- 5 **Moonshot 10×** — Push the best branch to its boldest form — SLIDES 5, 8
- 6 **Hard empathy** — Live one day inside each world, in 5 steps — SLIDES 5 to 7, 13
- 7 **Working backwards** — Plan right to left, from 2041 to today — SLIDES 14, 15

NEXT: THE ENGINE'S FIRST OUTPUT: THE BEST WORLD WE CAN IMAGINE.

TWO CRITICAL UNCERTAINTIES SET THE AXES:

AI shared and verified ↔ AI concentrated and opaque; Energy scarce, polluting ↔ Energy abundant, clean. **Symbiosis · Patchwork · Overheat**

AI "bad copying" — We asked AI for dozens of variants of each world and kept the strange ones. Its mistakes became raw material.

Exaptation — Borrowed from air traffic control (one network, many vehicles) and multiplayer game design (rules that shape behaviour).

A future we want to be part of

MOONSHOT THINKING: THE UPWARD WHAT IF — SHARED, VERIFIED AI WITH ABUNDANT CLEAN ENERGY

Mirror Cities — Customers try a flat or a lunar hotel in a live digital twin before buying. White-hat Octalysis: each purchase plants a real tree.

Life and transport — Electric pods and air taxis on one AI network, borrowed from air traffic control. Car parks become parks.

Space and the Moon — A crewed base at the lunar south pole mines ice for fuel. Robots build; the first tourists arrive.

Our industry — AI writes the analysis; consultants own the judgement. E, O&T move: launch a future-design practice.

A DAY IN 2041 | HARD EMPATHY IN 5 STEPS — LAYLA, 34, GRID ENGINEER, RIYADH

Scene: 6 am. My AI agent has already sold my solar surplus and drafted my report.

Feel: free, and strangely idle.

Make sense: the machines now do the routine half of my job.

Worry: what am I still for?

Act: I join the council that designs our city's digital twin.

5
CS

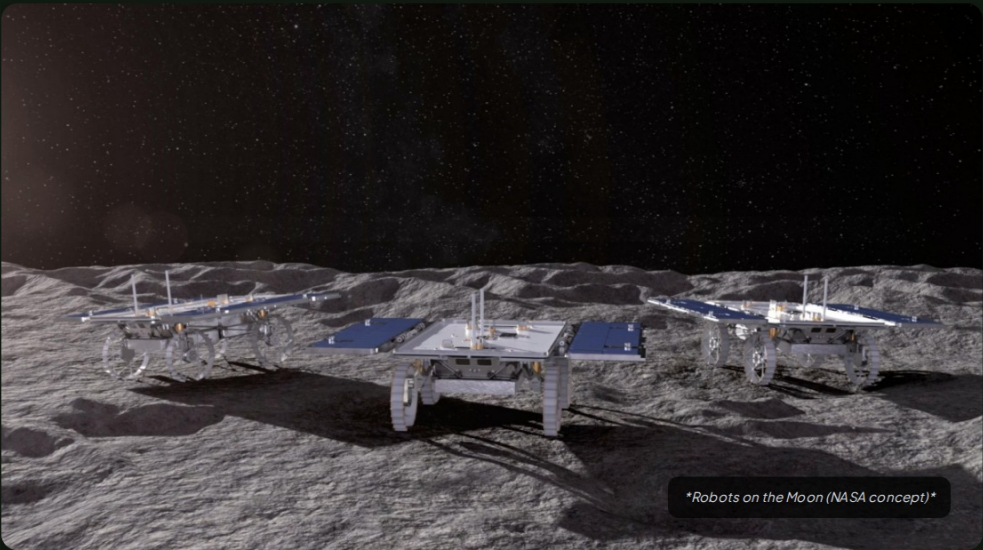
CHARACTER: LAYLA,
AND EVERY CLIENT LIKE
HER

CONFLICT: AI'S
HUNGER FOR
ENERGY

CURE: CLEAN
POWER, EFFICIENT
CHIPS

CHANGE:
ABUNDANCE,
SHARED WIDELY

CARRY-OUT: BUILD
THE FUTURE WE
WANT



Every utopia has a shadow

The honest shadow of World 1. Same engine, downward What if, told as a chain of cause and effect.

Counterfactual: the downward What if — Concentrated, opaque AI with scarce, polluting energy

HOW IT HAPPENED, ONE "BECAUSE OF THAT" AT A TIME

1. AI demand outran clean power, so fossil plants kept running.
2. Heat and pollution rose, and power was rationed.
3. A few players owned the models; deepfakes broke trust.
4. A debris cascade took out navigation and broadband satellites.

Customers — The metaverse is an escape; black-hat Octalysis (scarcity, loss) keeps people hooked.

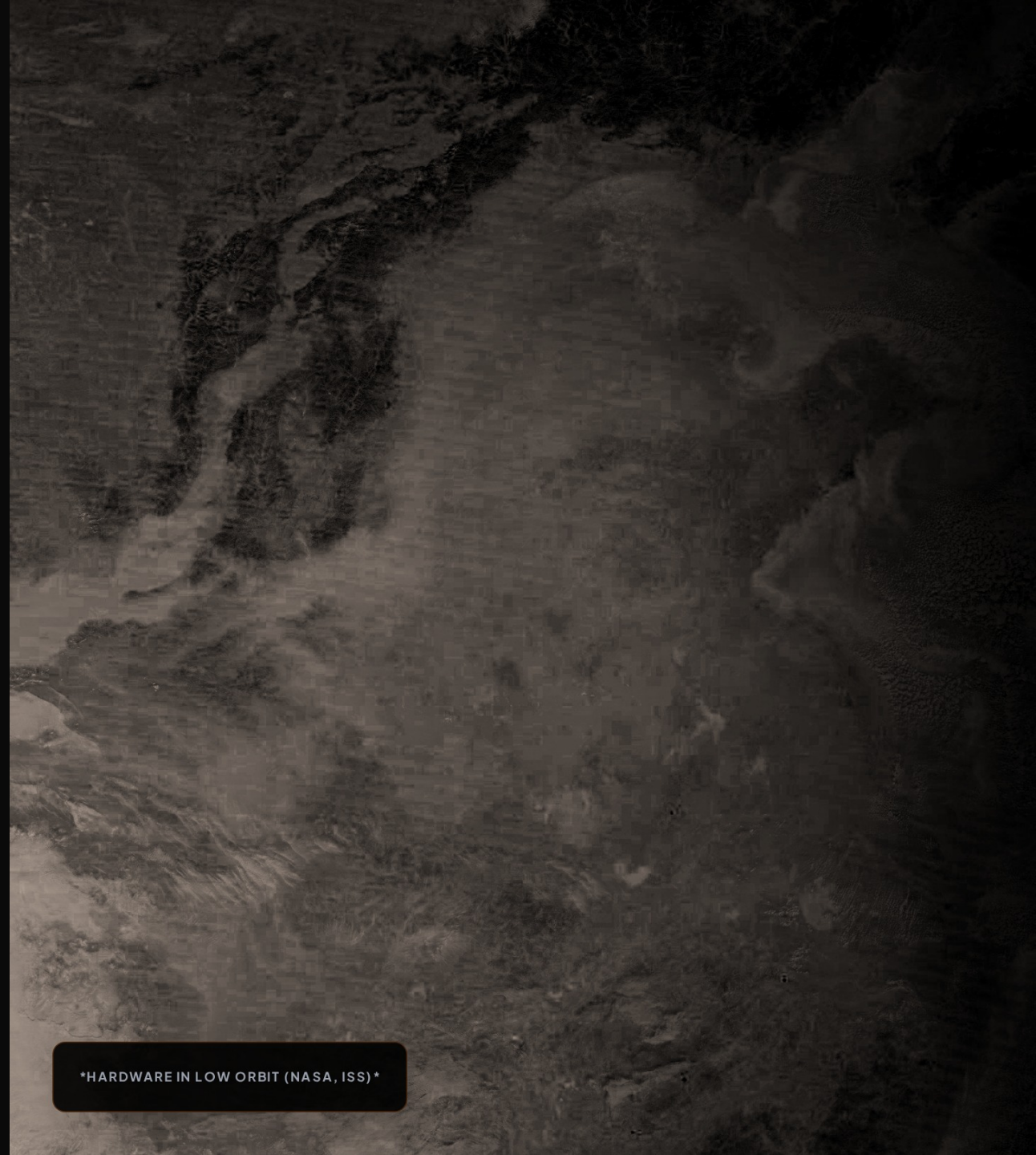
Space and the Moon — Low orbit is a junkyard; the lunar base waits.

Life and transport — Empty robotaxis gridlock the cities. The rich fly; everyone else waits.

Our industry — Reports cost pennies. E,O&T move: verification and resilience services.

A DAY IN 2041 | HARD EMPATHY IN 5 STEPS — OMAR, 41, LOGISTICS MANAGER

Scene: in the depot, two delivery alerts contradict each other. · Feel: anxious; the rationing app also limits when I can charge the vans. · Make sense: one of the alerts may be synthetic. · Worry: I could send a driver onto an unsafe route. · Act: I verify both alerts through our data cooperative before dispatch.



Neither heaven nor collapse

The middle path, built by flipping two of today's facts with Tick-Tock. Here trust, not technology, decides who wins. **Tick-Tock: the middle path** — Trust verified inside regional blocs, mixed energy between them. *One planet, three rulebooks.*

TICK People buy from companies.

→ TOCK AI agents buy from AI agents, so firms must design for machine customers.

TICK Data is free.

→ TOCK Data is owned and paid for, like energy.

Customers — Regional metaverses joined by one digital identity, so trust becomes the product.

Our industry — Winners translate between blocs. E,O&T move: a cross-bloc standards advisory.

Space and the Moon — Shared on paper; blocs compete quietly for lunar ice and landing sites.

Life and transport — Driverless in some regions, manual in others; the best cities blend both.



A DAY IN 2041 | HARD EMPATHY IN 5 STEPS — SARA, 29, PRODUCT DESIGNER, STOCKHOLM

"Scene: my AI agent is doing my weekly shopping. · Feel: relaxed, until it refuses my favourite app. · Make sense: that app has no trust seal in our bloc. · Worry: my own product could be locked out the same way. · Act: I design one product that passes all three blocs' trust rules."

From your door to orbit

ZOOM IN (1) | MOONSHOT AND EXAPTATION · STORY SPINE: BECAUSE OF THAT: ZOOM IN

Borrow the delivery-network logic, not the aircraft: a model that brings dinner today, adapted for orbital logistics, could help feed astronauts in 2041.



2026 · OBSERVED

Drones deliver food now. One delivery network passed 2 million commercial drone deliveries in January 2026, with a median flight of 3 minutes (Food On Demand). TRANSFER: scheduling, tracking, exception handling



2030s · PROPOSED TEST

Autonomy reaches orbit. Robotic freighters already berth at the station using its robotic arm. Next: test AI scheduling with qualified space partners. GATE: safety, communication delay, payload limits



2041 · MOONSHOT

Feeding the crew in space. Each astronaut needs about 1.7 kg of food a day; fresh produce arrives every three months (NASA). Imagine cargo and orbital crops supporting a lunar outpost. 10x AMBITION: on-demand, not occasional, resupply

E,O&T test. Start with a logistics simulation and a partner review, not launch infrastructure. For clients in logistics and food, the last mile becomes the last orbit.

NEXT: ZOOM IN: WHEN GOLD STOPS BEING THE TREASURE.

The new gold

ZOOM IN (2) | TICK-TOCK AND ALL FOUR LENSES · STORY SPINE: BECAUSE OF THAT: ZOOM IN

TICK Gold is the treasure. **TOCK** Water, air, sand and plants are.

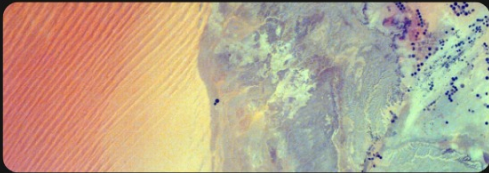
In 2041 value may sit in what keeps systems alive. Each card applies one of the 4 lenses of innovation and ends with an E,O&T opportunity.

And gold? Still useful: it coats the mirrors of the Webb space telescope. But in 2041 it no longer measures wealth.



WATER

Training GPT-3 was estimated to evaporate about 700,000 litres of fresh water (Li et al.). The space station recycles 98% of its water (NASA). **LEVERAGE RESOURCES:** Adapt closed-loop thinking to a client water audit.



SAND

Sand and gravel use is about 50 billion tonnes a year, 17 kg per person per day (UNEP). Desert sand is too round for concrete. **CHALLENGE** **ORTHODOXIES:** Replace "abundant means disposable" with reuse by design.



AIR

Lunar soil is 40 to 45% oxygen by weight (ESA), but it is chemically bound and not breathable until extracted. **HARNESS TRENDS:** Track extraction tests, energy needs and readiness.



PLANTS

Lettuce and chile peppers grow on the station (NASA): crop research, not yet a complete food system. **UNDERSTAND NEEDS:** Design for nutrition, reliable supply and the crew's experience.

NEXT: THREE WORLDS, ONE TREE. WHICH BRANCH ARE WE ON?

Which branch are we on?

POSSIBILITY TREE | CONTINGENCY PLANNING · STORY SPINE: UNTIL FINALLY

Three worlds with planning weights, then one key driver per world split into three exclusive outcomes. Conditional odds sum to 100% per world, and the joint odds of all nine outcomes sum to 100%. Illustrative judgements, not calibrated forecasts.



How to read the tree

Level 1: worlds — Planning weights for the three worlds: 30%, 45%, 25%. They sum to 100%.

Level 2: one driver each — Three mutually exclusive outcomes per world, with conditional odds that sum to 100% within that world.

Joint odds — World x outcome. Worked example: Patchwork and partly compatible = 45% x 40% = 18%. All nine joint odds total 100%.

Why Patchwork leads — AI rules are already splitting by region; the clean-power race is not yet won.

STATUS: ILLUSTRATIVE JUDGEMENTS TO SUPPORT DISCUSSION, NOT CALIBRATED FORECASTS.

NEXT: A LIGHT ONLY MATTERS IF IT CHANGES A DECISION.

A light only matters if it changes a decision

WATCH | EARLY-WARNING LIGHTS THAT CHANGE ACTION · STORY SPINE: UNTIL FINALLY: WATCH

Each signpost has a record, a review trigger and a response. Triggers are proposed review rules measured against a pilot baseline, not external benchmarks.

SIGNPOST	WHAT WE RECORD	REVIEW TRIGGER	RESPONSE	POINTS TO
Clean power for AI	Clean energy built per new data centre; energy used per completed task	Two monthly reviews worse than baseline	Switch to a smaller model or a manual route	OVERHEAT
Evidence quality	Share of outputs with a traceable source; material errors	Any unmitigated material error	Pause automation; require human approval	SYMBIOSIS OR OVERHEAT
Portability	Time and failures when moving one test case between providers	A second provider fails the same test	Add an adapter; avoid single-provider lock-in	PATCHWORK
Orbital access	Debris-removal missions flown; launch delays	Removal still experimental by 2030	Plan ground fallbacks for satellite services	OVERHEAT
Human agency	Can users understand, challenge and reverse a result?	A high-impact decision has no appeal route	Redesign the handover before scaling	ALL THREE WORLDS

Sell better decisions, not more reports

RESPOND | THE E,O&T BUSINESS MODEL · STORY SPINE: AND EVER SINCE THEN

Three proposed offers, one per world, share one reusable asset: a Decision Passport that records why a client should act, who owns the decision and what happens if it fails.

FUTURE DESIGN LAB

CLIENT Strategy and innovation leaders. Rehearse a new service in a digital twin before committing. VALUE TEST Time to a testable decision

VERIFIED DECISIONS AND CONTINUITY

CLIENT Operations and risk leaders. Check evidence, rehearse an outage and prove a fallback. VALUE TEST Material errors; fallbacks completed

INTEROPERABILITY STUDIO

CLIENT Product and technology leaders. Separate the core service from provider-specific adapters. VALUE TEST Successful transfers; switching effort

NEXT: ONE WORLD, ONE PERSON, THE FULL STORY SPINE IN EIGHT STEPS.

DECISION PASSPORT

WORKED EXAMPLE: OMAR'S DISPATCH IN OVERHEAT (ILLUSTRATIVE REHEARSAL)

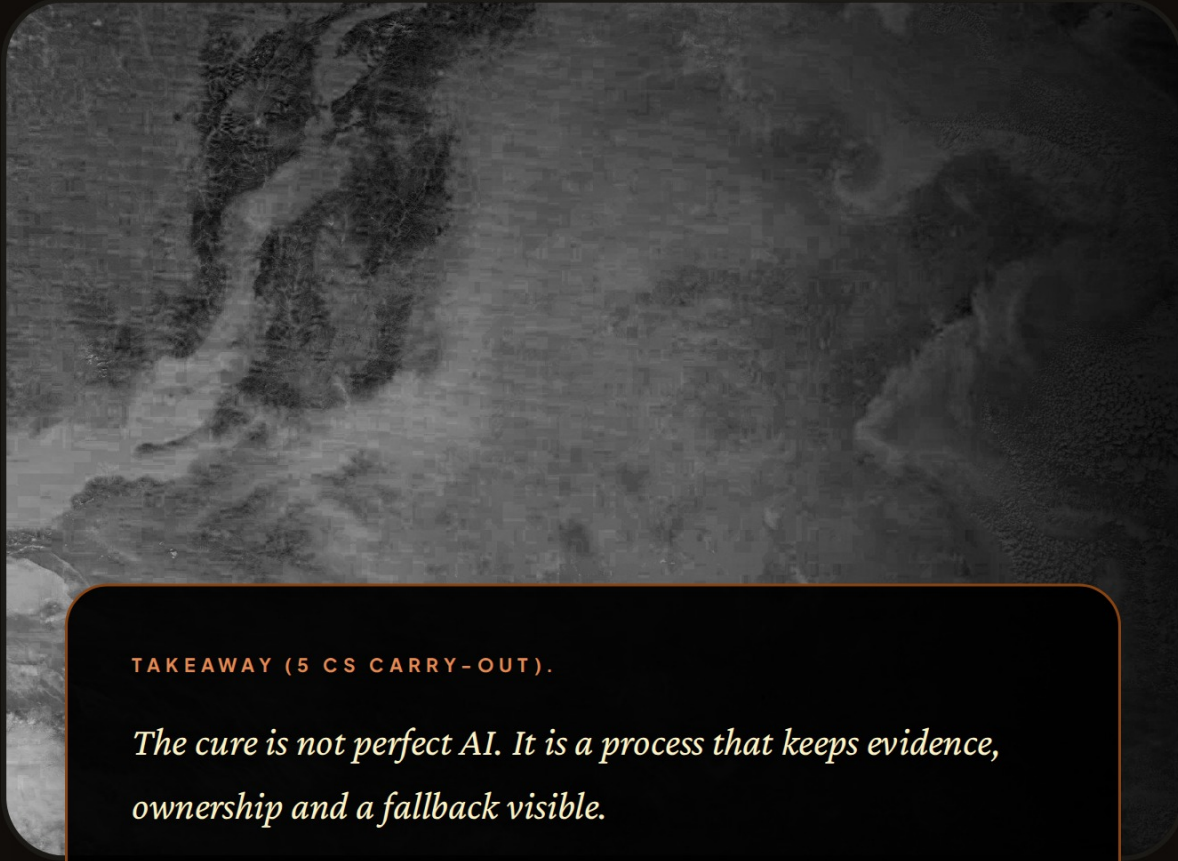
DECISION	May the AI dispatch this van, or must Omar use the verified manual route?
ASSUMPTION	Route and charging data are current enough for this dispatch.
EVIDENCE	Time-stamped sources, a human route check and pilot results, still to be collected.
OWNER	Omar authorises; dispatchers can flag an alert.
REVIEW SIGNAL	Conflicting or untraceable alerts, or no charging available.
FALLBACK	Verify by phone, use the manual route and tell the customer.

Proposed service concepts and an illustrative rehearsal, not existing products or results.

One causal story inside one world

STORYTELLING | THE STORY SPINE, STEP BY STEP · STORY SPINE: STORY IN ONE WORLD

The worlds branch, but a story inside a world must follow cause and effect. All eight Story Spine steps for Omar in Overheat, a fictional rehearsal.



TAKEAWAY (5 CS CARRY-OUT).

The cure is not perfect AI. It is a process that keeps evidence, ownership and a fallback visible.

- 1 **ONCE UPON A TIME** — Omar ran a depot whose customers depended on daily deliveries.
- 2 **EVERY DAY** — His team dispatched the vans on the AI's recommendation.
- 3 **BUT ONE DAY** — **Two alerts conflicted, and the planned van could not charge.**
- 4 **BECAUSE OF THAT** — Omar paused the recommendation before sending a driver.
- 5 **BECAUSE OF THAT** — He checked the records and called the depot to verify the route.
- 6 **BECAUSE OF THAT** — He used the manual fallback and logged why the decision changed.
- 7 **UNTIL FINALLY** — **A verified alternative kept the delivery moving.**
- 8 **AND EVER SINCE THEN** — The team reviews every exception with a Decision Passport.

NEXT: FROM ONE STORY TO ONE PLAN: WORK BACKWARDS FROM 2041.

From 2041 back to Monday morning

WORKING BACKWARDS | RIGHT TO LEFT · STORY SPINE: AND EVER SINCE THEN

How the worlds become action. We start at the end in 2041 and plan step by step, right to left, so every move made today is useful in any world.

4

2026

NO-REGRET MOVES THIS YEAR

A digital-twin pilot with one client. A signpost dashboard that tracks the early-warning lights. A quarterly "day in 2041" hard-empathy workshop with each client team.

3

2030

ENERGY-AWARE AI

Every client plans AI around its power footprint, and reskilling covers the 39% of skills expected to change.

2

2035

MACHINE CUSTOMERS, VERIFIED SUPPLY

Each client runs a strategy for AI agents as buyers and a verified AI supply chain.

1

2041 START HERE

PRESS RELEASE FROM THE FUTURE

"E,O&T clients thrive in whichever world arrived."

Trusted AI, clean power and a usable orbit.

CREATIVE FUEL FOR THE TEAM

A shared commonplace book captures every signal and slow hunch; a planned stimuli queue (AI 2041, Imaginable, one sci-fi film a month) feeds the next round of scenarios.

NEXT: THE FIRST 90 DAYS: PROVE IT IS USEFUL BEFORE SCALING IT.

Prove it is useful before scaling it

FIRST 90 DAYS | SMALL TESTS, CLEAR OWNERS · STORY SPINE: EVER SINCE: FIRST TEST

One client, one workflow, one limited rehearsal. All activities and gates below are proposed; no client commitment or benefit is claimed.

01

DAYS 1 TO 30 — DEFINE AND OBSERVE

OWNERS Client owner and consultant. Map one workflow, record a manual baseline and interview five users. **GATE:** agree success, error severity, data permissions and limits.

02

DAYS 31 TO 60 — PROTOTYPE AND REHEARSE

OWNERS Technical lead and client users. Build a simulation and a Decision Passport; rehearse all three worlds. **GATE:** five users can name the evidence, owner and fallback.

03

DAYS 61 TO 90 — STRESS-TEST AND DECIDE

OWNERS Sponsor and independent reviewer. Run ten scripted cases: low power, false input, provider outage and normal operation. **GATE:** every critical fallback works; no unresolved material error.

CONTINUE, CHANGE OR STOP

Scale only when the pilot shows client-valued benefit and acceptable resource use.

PILOT LIMIT. FIVE USERS AND TEN CASES SUPPORT LEARNING, NOT PROOF OF GENERAL RELIABILITY.

NEXT: KEEP THE TEAM CREATIVE LONG AFTER THIS BRIEFING.



Rehearse, reflect, feed the next idea

KEEP CREATING | A REPEATABLE 45-MINUTE ROUTINE · STORY SPINE: EVER SINCE: KEEP CREATING

A proposed monthly workshop that links the creative conditions, motivating feedback and future-thinking tools from this briefing.

01 GENERATE | 10 MIN — A SAFE ROOM FOR ALTERNATIVES

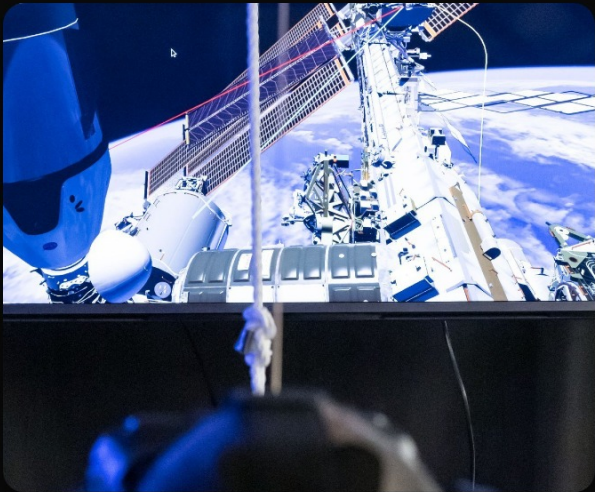
PSYCHOLOGICAL SAFETY. Write ideas silently, then share. The leader invites disagreement; challenge the assumption, not the person.

02 REHEARSE | 20 MIN — LEARN THROUGH THE SCENARIO

OCTALYSIS, USED ETHICALLY. Pick a world and a role. A shared progress bar tracks evidence, owner and fallback; retries are allowed after feedback.

03 REFLECT | 15 MIN — KEEP THE USEFUL SURPRISE

AI AND SCIENCE FICTION. Try one strange AI or sci-fi variation as an idea, not a fact. Check its assumptions and keep a testable version in the commonplace book.



The Moon is closer than ever. So is 2041.

CLOSING STORY SPINE: CARRY-OUT MESSAGE

Symbiosis **Patchwork** **Overheat**

BUILD IT

SHAPE IT

INSURE
AGAINST IT

THREE WORLDS, ONE DISCIPLINE: MAKE THE ASSUMPTIONS VISIBLE, TEST
THE RESPONSE, AND KEEP A HUMAN DECISION OWNER IN THE LOOP.

"We do not predict the future. We prepare for all three, and we work every day for the first one."



Mohammad

SENIOR FUTURES CONSULTANT, EM, OLI & TIMMONDS

NEXT: PART 2, MY PERSONAL REFLECTION ON THE TOOLS BEHIND THIS BRIEFING.

What I am taking with me

PART 2 | PERSONAL REFLECTION | 1 OF 2

The tools that resonated, the change in my thinking, and why it matters in my professional life.

Three ideas from this course changed how I think about planning: possibility trees, working backwards, and counterfactual thinking. Together they gave me a structured language for something my work has always demanded: decisions about a future nobody can see.

In twenty years of network engineering and digital transformation, I have learned that a platform chosen today may still shape a business in ten years. Building the three worlds of 2041 for this assignment showed me a different way to approach that kind of decision. Drawing the possibility tree forced me to give each branch a probability, and the less likely branches raised questions a single forecast would never have asked.

Working backwards from a 2041 press release, the way Amazon writes the press release before the product exists, made the first steps for 2026 obvious. Counterfactual thinking kept the exercise honest: writing Overheat next to Symbiosis stopped me from falling in love with the optimistic future.

20 years

IN NETWORKS AND DIGITAL
TRANSFORMATION

where decisions have long consequences



"What resonated most is that creativity here is not a talent but a structured skill, wild ideas plus strategy."

EM, OLI & TIMMONDS | THREE WORLDS OF 2041 -
REFLECTION: 334 WORDS ACROSS SLIDES 18 AND 19

How I will put it into practice

PART 2 | PERSONAL REFLECTION | 2 OF 2

A realistic timetable, a review loop and a personal routine that continues beyond this course.

My plan is concrete. From November, every new technical proposal I write will open with a one-page working backwards brief describing the result for the customer in three years, the assumptions behind it and the first test.

Once a quarter, starting in January 2027, I will build a possibility tree for one roadmap decision, with three branches, nine sub-branches and probabilities, and review it three months later to see which branch reality followed. Before each major design review I will run one upward and one downward counterfactual with the team and record the risks it surfaces.

Personally, I live between Riyadh and Stockholm, and I will turn those flights into fuzzy focus time: no screens, just my stimuli queue and a commonplace book, so slow hunches have somewhere to land.

NOV 2026

Working-backwards briefs

JAN 2027

First quarterly tree

APR 2027

Review 1 and tree 2

JUN 2027

Assess decision impact

"I will measure success simply: by June 2027, I will have built two possibility trees, completed the first three-month review, and recorded whether the findings changed a roadmap decision."

SOURCES & USE OF AI

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IMAGE CREDITS

Slide 1: "A Breathtaking Earthset from Orion", Artemis II crew, 6 April 2026. NASA (art002e021278).

Slide 3: Night Earth observation from the ISS. Expedition 40. NASA (iss040e085126).

Slides 2, 4 and 14: "Home in the Distance", Artemis II. NASA (art002e020922).

Slide 5: "Earth in View", Artemis II (art002e023506); CADRE rovers concept, NASA/JPL-Caltech (PIA26296).

Slide 6: "Haze over eastern China", NASA Goddard (GSFC_20171208_Archive_e001269); ISS Expedition 35 (iss035e017673).

Slides 7, 10 and 17: "Black Marble: Africa, Europe, and the Middle East", NASA Goddard (GSFC_20171208_Archive_e001591).

Slides 5 to 7 and 16: NASA VR training (jsc2026e000427), Advanced Air Mobility concepts (AAM-Concept-2021, automation concept), Mission Control (jsc2026e019925), Pheleas supercomputer (ARC-2008-ACD08-0272-005).

Slide 8: NASA drone flight test (AFRC2026-0152-22); Cygnus in Canadarm2 (iss071e418235); Expedition 66 chile peppers (iss066e023179).

Slide 9: ISS water recovery (iss072e143163); sand dunes and irrigation in Saudi Arabia (PIA11069); Earth limb and Moon (iss051e039977); Veg-01 harvest (iss040e009126); JWST golden mirror (GSFC_20171208_Archive_e000368).

World thumbnails on slides 2, 4, 10, 12, 13 and 17 reuse the images of slides 5, 6 and 7.

All NASA images via the NASA Image and Video Library. <https://images.nasa.gov/public-domain>.

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AI DECLARATION

I acknowledge the use of Claude by Anthropic (<https://claude.ai>) and ChatGPT by OpenAI (<https://chatgpt.com>) to help generate and develop scenario ideas and variants, critique and develop wording, framework applications, scenario logic and probabilities, design the Decision Passport, pilot and workshop, find and check sources, and arrange and format the slides.

The prompts used include "Propose solution directions for each company and the course concepts they would apply", "Draft three different future scenarios for 2041 with personas, built on the course tools", "Check every slide against the assignment requirements and rubric", "Merge the strongest ideas from my two drafts into one deck" and "Build the deck as a PowerPoint with my photo and a dark cinematic layout".

The output was used as a starting point that I selected, reviewed, edited and verified; the choice of company and direction, the final scenarios, and the reflection based on my own experience and plans are mine, and I remain responsible for the submission.

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